Microbes in Human Welfare

Complete Chapter Tutorial, Structural Diagrams, Comparative Tables, and NEET-Aligned Practice Questions with Detailed Explanations

Learning Objectives & Exam Weightage

This chapter constitutes an essential scoring section of Applied Biology for the NEET UG examination, regularly accounting for 1 to 3 direct multiple-choice questions (4 to 12 marks). By the conclusion of this tutorial, learners will master:

- **Household Fermentations:** The role of Lactic Acid Bacteria (LAB), starter inoculum, vitamin enrichment, and yeast fermentations in traditional foods.
- **Industrial Applications:** Fermentor technology, distilled vs non-distilled alcoholic beverages, penicillin discovery, and high-yield bioactive molecules (statins, cyclosporin A, streptokinase, and organic acids).
- **Sewage Treatment Plants (STP):** Two-tier treatment (primary physical filtration/sedimentation vs secondary biological oxidation via aerobic flocs), biochemical oxygen demand (BOD) dynamics, activated sludge, and anaerobic sludge digesters.
- **Biogas Production:** Methanogens (*Methanobacterium*), rumen cellulose digestion, floating gas-holder biogas plant architecture, and institutional contributions (IARI and KVIC).
- **Biocontrol Agents:** Integrated Pest Management (IPM), predatory insects (ladybird and dragonflies), *Bacillus thuringiensis* (Bt), *Trichoderma*, and narrow-spectrum, species-specific baculoviruses (*Nucleopolyhedrovirus*).
- **Biofertilisers:** Nitrogen fixation by symbiotic and free-living bacteria, mycorrhizal associations (genus *Glomus*), and cyanobacteria in wetland paddy cultivation.

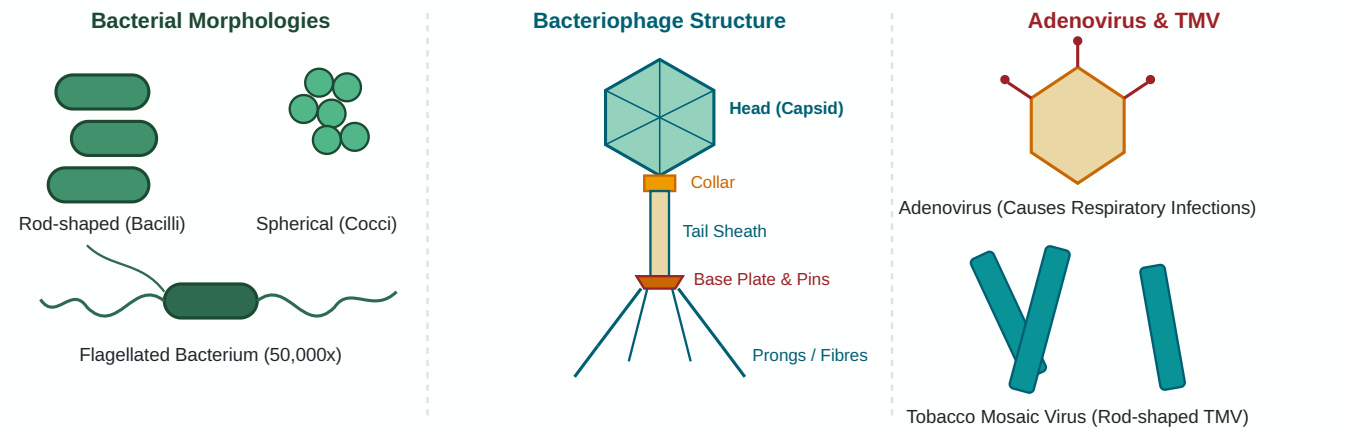
1. Overview of Microorganisms: Ubiquity and Diversity

Microorganisms represent major functional components of biological systems on Earth. Unlike macroscopic animals and plants, microbes colonize every possible ecological habitat: soil, water, ambient air, surfaces, and internal environments of animals and plants. Notably, extremophilic microorganisms thrive in inhospitable biotopes devoid of higher life forms, such as deep inside hydrothermal vents (geysers) where temperatures reach or exceed 100°C, deep subsoil strata, permafrost beneath snow layers several meters thick, and highly acidic water bodies.

Microbial diversity spans multiple biological domains and non-cellular entities:

- **Cellular Microbes:** Bacteria (Kingdom Monera), Protozoans (Kingdom Protista), and Microscopic Fungi (Kingdom Fungi). Bacteria and filamentous fungi can be cultivated on sterile nutritive agar media in Petri dishes to form macroscopic colonies that can be directly observed and quantified.
- **Acellular Infectious Entities:** Viruses (obligate intracellular nucleoprotein complexes), Viroids (free infectious low-molecular-weight RNA devoid of a protein capsid), and Prions (abnormally folded proteinaceous infectious agents capable of transmitting neurodegenerative conditions).

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Morphological diversity of representative microorganisms, acellular bacteriophage architecture, adenovirus, and rod-shaped tobacco mosaic virus (TMV).

2. Microorganisms in Household Products

Everyday domestic culinary practices rely on controlled fermentations driven by specialized bacteria and fungi.

2.1 Production of Curd and Lactic Acid Bacteria (LAB)

Fresh milk is converted into curd through the biological agency of **Lactic Acid Bacteria (LAB)**, predominantly comprising members of the genus *Lactobacillus* and related taxa.

- **Inoculation:** A small aliquot of curd, referred to as the **inoculum** or **starter**, is introduced into lukewarm fresh milk. This inoculum contains millions of viable LAB cells.
- **Biochemical Mechanism:** Under ambient physiological temperatures, LAB proliferate and metabolize lactose (milk carbohydrate) into lactic acid. The accumulated acid decreases the pH, causing the **coagulation and partial enzymatic digestion of milk proteins** (casein).
- **Nutritional Enrichment:** The fermentative activity of LAB improves the nutritional profile of milk by substantially increasing the concentration of **vitamin B_{12}** (cyanocobalamin).
- **Protective Probiotic Action:** Within the human gastrointestinal tract, established populations of LAB play a critical protective role by suppressing and checking the proliferation of opportunistic and pathogenic enterobacteria.

2.2 Fermentation of Dough for Traditional Foods

- **Dosa and Idli Dough:** The batter prepared from rice and black gram (Bengal gram flour or legumes) undergoes spontaneous bacterial fermentation. The characteristic puffed-up, porous texture of the leavening batter is driven by the internal entrapment of **carbon dioxide (CO_2)** generated as a byproduct of anaerobic fermentation.
- **Bread Dough and Baker's Yeast:** Dough utilized in bread production is fermented using the single-celled budding yeast *Saccharomyces cerevisiae*, commercially designated as **baker's yeast**. Glycolytic conversion of maltose/glucose produces ethanol and abundant CO_2, causing the spongy rise of the loaf during baking.

2.3 Traditional Drinks and Fermented Delicacies

- **Toddy:** A traditional indigenous beverage consumed in parts of southern India, prepared by tapping and fermenting the nutrient-rich sap harvested from palms using wild yeasts and bacteria.
- **Regional Fermentations:** Controlled microbial fermentation is conventionally utilized in processing perishable foodstuffs, including **fish**, **soybeans** (to yield miso and tempeh), and tender **bamboo shoots**.

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2.4 Cheese Manufacture: Role of Distinct Microbes

Cheese represents one of the earliest microbially processed food commodities. Unique varieties are characterized by specific textures, aromas, ripening periods, and flavors, directly dictated by the specific inoculum:

- **Swiss Cheese:** Renowned for its large macroscopic eyes or holes. These large gas pockets are formed by copious amounts of CO_2 synthesized during fermentation by the bacterium ***Propionibacterium freudenreichii* subsp. *shermanii*** (referenced in text as *Propionibacterium sharmani*), which metabolizes lactic acid into propionic acid, acetic acid, and carbon dioxide.
- **Roquefort Cheese:** A semi-hard, blue-veined cheese ripened through the invasive growth of a specific filamentous fungal mould, ***Penicillium roqueforti*** (a specific fungus), which imparts a distinct pungent aroma, texture, and flavor profile.

> **NEET High-Yield Trap: Microbe Identity in Cheeses**
>
> Do not confuse the microbial kingdom responsible for specific cheese varieties: **Swiss cheese** is produced by a *bacterium* (*Propionibacterium sharmani*, resulting in large CO_2 holes), whereas **Roquefort cheese** is ripened by a *fungus* (giving it specific flavor).

3. Microbes in Industrial Products

The industrial synthesis of microbially derived metabolites, alcoholic beverages, and pharmaceutical antibiotics requires the mass cultivation of pure cultures in high-volume, closed-containment vessels engineered with temperature, pH, and agitation controls, termed **fermentors** or bioreactors.

3.1 Fermented Alcoholic Beverages

The identical budding yeast species, *Saccharomyces cerevisiae*, used in baking is employed in distilleries and breweries under the title **brewer's yeast**. The yeast ferments fermentable sugars derived from malted cereals (barley) and crushed fruit juices into ethyl alcohol (C_2H_5OH) and CO_2:

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast Zymase Complex}} 2\,C_2H_5OH + 2\,CO_2$$

The commercial categorization of alcoholic beverages depends strictly upon two technical parameters: the nature of the organic carbohydrate substrate, and whether downstream processing includes thermal distillation:

Category	Representative Beverages	Downstream Distillation	Alcoholic Strength (%)
Undistilled Beverages	Wine (fermented grape juice), Beer (fermented malted barley)	**Produced WITHOUT distillation**	Low to moderate (approx. 3%–12%) due to natural yeast toxicity limit
Distilled Beverages	Whisky (malted grain mash), Brandy (distilled wine), Rum (fermented molasses)	**Produced WITH distillation** of the fermented broth	High (approx. 40%–55%), concentrated via fractional vapor recovery

3.2 Antibiotics

Antibiotics represent low-molecular-weight chemical substances synthesized by certain microorganisms that possess the capability to selectively kill (bactericidal) or retard the growth (bacteriostatic) of competing pathogenic organisms. Etymologically, *anti* translates to "against" and *bios* signifies "life" (denoting antagonism against disease-causing parasites); however, relative to human patients, antibiotics are strictly **pro-life**.

Historical Milestone: The Discovery of Penicillin

- **Serendipitous Discovery:** In 1928, Scottish microbiologist **Alexander Fleming**, while investigating pathogenic *Staphylococci* bacteria, observed that an unwashed nutrient culture plate had developed a contaminating mould. Around this mould colony, adjacent colonies of *Staphylococci* underwent lysis and

failed to grow. Fleming established that the mould secreted a diffusable inhibitory bactericidal metabolite, which he christened **Penicillin** after the producing fungal mould, ***Penicillium notatum***.

- **Therapeutic Realization & Mass Production:** Fleming was unable to chemically stabilize and isolate penicillin in sufficient therapeutic purity. Its complete pharmacological efficacy as a systemic antibiotic was elucidated over a decade later by **Howard Florey** and **Ernst Chain**.
- **Wartime Deployment & Nobel Recognition:** Penicillin was deployed on an unprecedented scale during World War II to treat infected, wounded Allied soldiers. In 1945, **Fleming, Chain, and Florey were jointly awarded the Nobel Prize in Physiology or Medicine**.
- **Clinical Scope:** The development of penicillin and subsequent microbial antibiotics transformed modern therapeutics, curbing fatal bacterial epidemics such as **plague**, **whooping cough** (vernacular: *kali khansi*), **diphtheria** (vernacular: *gal ghotu*), and **leprosy** (vernacular: *kusht rog*).

3.3 Organic Acids, Enzymes, and Bioactive Compounds

Industrial bioprocessing exploits microbial strains for high-yield synthesis of organic acids, commercial enzymes, and bioactive pharmaceutical molecules:

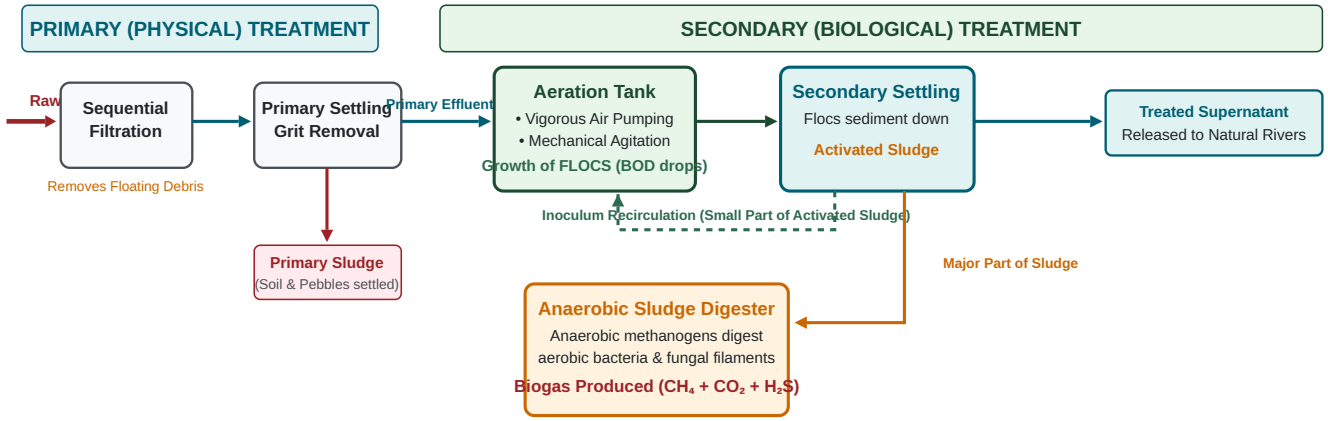
Class	Product Molecule	Producing Microorganism	Microbial Group	Commercial / Clinical Function
Organic Acids	Citric acid	*Aspergillus niger*	Fungus (Ascomycete mould)	Food preservative, flavoring, and industrial buffering agent
	Acetic acid	*Acetobacter aceti*	Bacterium	Vinegar synthesis, industrial chemical precursor
	Butyric acid	*Clostridium butylicum*	Bacterium (Anaerobic)	Ester synthesis, chemical feedstock
	Lactic acid	*Lactobacillus*	Bacterium	Food preservation, tanning, acidification
Enzymes	Lipases	Various fungal/bacterial strains	Microbial enzymes	Detergent formulations; removes hydrophobic **oily stains** from fabrics
	Pectinases & Proteases	Fungal and bacterial cultures	Microbial enzymes	**Clarification of commercial bottled fruit juices**, preventing turbidity
	Streptokinase	*Streptococcus* (genetically engineered strains)	Bacterium	**"Clot buster"**; lyses intravascular thrombi in patients with acute myocardial infarction (heart attack)
Bioactive Molecules	Cyclosporin A	*Trichoderma polysporum*	Fungus	Potent **immunosuppressive agent** administered to prevent graft rejection in organ transplant recipients
	Statins	*Monascus purpureus*	Yeast (Fungus)	**Blood-cholesterol lowering agent**; acts via competitive inhibition of the rate-limiting enzyme in cholesterol biosynthesis (HMG-CoA reductase)

4. Microbes in Sewage Treatment Plants (STPs)

Municipal wastewater generated continuously across human urban settlements is designated as **sewage**. Sewage is loaded with substantial quantities of organic putrescible matter and human excreta, and harbors dense populations of pathogenic enteric parasites, viruses, and bacteria. Raw sewage cannot be discharged into freshwater bodies (rivers, lakes, and streams) without precipitating severe eutrophication, oxygen collapse, and epidemics of water-borne diseases.

Domestic wastewater treatment is accomplished inside municipal **Sewage Treatment Plants (STPs)** through the metabolic agency of diverse **heterotrophic microbes** naturally resident in the sewage itself. STP operations are executed in two consecutive, distinct structural stages:

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Comprehensive flowchart of Sewage Treatment Plant (STP) architecture detailing Primary Physical Treatment vs Secondary Biological Treatment, Activated Sludge recirculation, and Anaerobic Sludge Digestion producing inflammable Biogas.

4.1 Primary Treatment (Physical Separation)

Primary treatment consists strictly of **physical operations** intended to remove suspended and settleable solids across sequential steps:

- **Sequential Filtration:** Coarse and fine floating debris is trapped and removed mechanically by passage through wire-mesh screens of progressively smaller aperture.
- **Sedimentation:** The filtered fluid enters grit chambers where mineral particles, such as soil, silt, sand, and small pebbles (collectively termed **grit**), sediment under gravitational settling.
- **Primary Sludge vs Effluent:** All solids that settle to the floor of the primary settling chamber constitute the **primary sludge**. The unclarified, organic-laden supernatant liquid forms the **primary effluent**, which is transferred downstream into secondary biological units.

4.2 Secondary Treatment (Biological Oxidation)

Secondary treatment constitutes a true **biological process** driven by aerobic heterotrophic organisms:

- **Aeration Tank Dynamics:** Primary effluent is channeled into massive open-air aeration tanks equipped with continuous mechanical surface aerators and compressed air injection spargers. Agitation saturates the liquid with molecular oxygen, facilitating the explosive proliferation of aerobic heterotrophic microorganisms into **flocs**.
- **Definition of Flocs:** Flocs are dense, macroscopic, mesh-like structural networks formed by vast **masses of aerobic bacteria interwoven with branching fungal mycelial filaments**.
- **Reduction of Biochemical Oxygen Demand (BOD):** As the flocs rapidly multiply, their vigorous respiration breaks down and metabolizes the dissolved and colloidal organic compounds present in the effluent. This intensive microbial consumption drives a steep, measurable decline in the effluent's **Biochemical Oxygen Demand (BOD)**.
- **BOD as an Indirect Metric:** *BOD expresses the precise quantity of molecular oxygen in milligrams* (mg) *that would be consumed by microorganisms to completely oxidise and decompose all organic substrates present in one liter of wastewater.* Therefore, **BOD is an indirect quantitative index of organic pollutant load**: higher BOD signifies greater organic putrescibility and elevated polluting potential. The biological aeration process is continued until the BOD drops to negligible levels.

4.3 Settling Tank, Activated Sludge, and Anaerobic Digestion

- **Secondary Settling & Activated Sludge:** When the BOD is sufficiently reduced, the mixed liquor flows into the secondary settling (clarifier) tank. Here, agitation ceases, allowing the bacterial-fungal flocs to gravitationally sediment. This settled biological mass is termed **activated sludge**.

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- **Inoculum Recycling:** A small fraction of the activated sludge is pumped back upstream into the aeration tank to serve as an active biological **inoculum**, sustaining the microbial population for incoming wastewater.
- **Anaerobic Sludge Digestion:** The remaining majority of the activated sludge is directed into closed, sealed chambers designated as **anaerobic sludge digesters**. In this strictly anoxic environment, specialized anaerobic bacteria proliferate and enzymatically lyse and digest the dead biomass of aerobic bacteria and fungi.
- **Biogas Generation:** During anaerobic sludge digestion, methanogens synthesize a gaseous fuel mixture termed **biogas**, comprising predominantly **methane** ($\mathrm{CH_4}$), along with **carbon dioxide** ($\mathrm{CO_2}$) and **hydrogen sulphide** ($\mathrm{H_2S}$). Because methane is inflammable, this gas can be captured as an on-site source of clean thermal and electrical energy.
- **Environmental Release:** The clarified supernatant effluent emerging from the secondary settling tanks, stripped of the bulk of its organic impurities and microbial pathogens, is safely discharged into receiving natural water bodies (rivers and streams).

4.4 River Action Plans

Accelerating demographic growth and unplanned industrial urbanization have vastly amplified the volume of municipal sewage. Because municipal STP construction has historically failed to match urban wastewater output, millions of liters of untreated municipal sewage are frequently dumped into natural drainages, provoking severe aquatic degradation, loss of biodiversity, and outbreaks of water-borne infections.

To rehabilitate India's vital aquatic arteries, the Ministry of Environment and Forests initiated extensive national conservation programs, notably the **Ganga Action Plan (GAP)** and the **Yamuna Action Plan (YAP)**. These plans fund the construction of large-capacity STPs to ensure that only biologically treated, non-polluting effluent is discharged into major rivers.

5. Microbes in Production of Biogas

Biogas is an inflammable gaseous fuel mixture predominantly containing **methane** ($\mathrm{CH_4}$), generated by microbial degradation of biomass under anaerobic conditions. The precise composition of gaseous end-products generated during microbial metabolism depends upon the microbial taxa involved and the specific biochemical nature of the organic substrates utilized.

5.1 Methanogens and Substrate Specificity

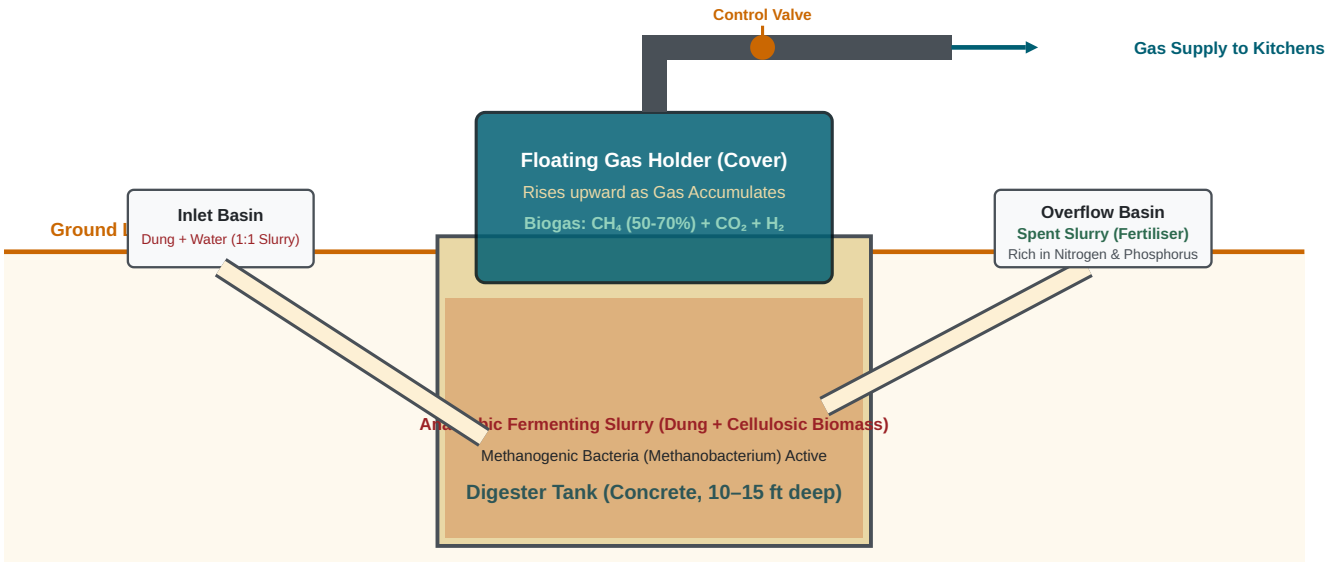
While carbohydrates fermented in leavening dough, cheese making, and alcohol brewing release $\mathrm{CO_2}$ as the primary gas, specialized obligate and facultative anaerobic bacteria that metabolize **cellulosic biomass** generate substantial quantities of $\mathrm{CH_4}$ along with $\mathrm{CO_2}$ and $\mathrm{H_2}$. These organisms are collectively designated as **methanogens**, with ***Methanobacterium*** representing a prominent model genus.

- **Natural Habitats:** Methanogens occur naturally in anoxic ecological niches:
  1. Inside the anaerobic sludge digesters of municipal sewage treatment plants.
  2. Within the **rumen** (the specialized forestomach chamber) of herbivorous ruminant livestock (cattle, buffaloes, goats).
  3. In water-logged marshy soils and anaerobic paddy sediments.
- **Ruminant Digestion:** Cellulosic plant forage ingested by cattle enters the rumen, where dense populations of methanogenic bacteria break down cellulose, releasing volatile fatty acids and microbial proteins essential for the animal's nutrition. Human digestive tracts lack the specialized cellulolytic symbionts required to digest dietary cellulose.
- **Gobar Gas:** Because cattle excreta (dung), vernacularly called *gobar*, contains an abundance of methanogenic bacteria along with undigested cellulosic plant fiber, it serves as the premier substrate for the production of biogas, commonly known as **gobar gas**.

5.2 Structural Design of a Rural Biogas Plant

The standard floating-cover biogas plant, widely deployed across rural India, operates on continuous anaerobic digestion:

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Architectural layout of a typical floating gas-holder biogas plant showing underground digester chamber, slurry inlet, floating steel gas collector, gas distribution line, and spent slurry fertilizer extraction.

- **Digester Pit:** A cylindrical concrete well, approximately **10 to 15 feet in depth**, built into the ground. It receives organic agricultural refuse, cattle dung, and water.
- **Slurry Preparation and Inlet:** Dung is blended with water in an above-ground mixing chamber to produce a uniform slurry, which is introduced via an inclined inlet pipe directly into the base of the digester tank.
- **Floating Gas Cover:** A mobile cylindrical steel or composite cover floats freely over the fermenting slurry. As methanogens metabolize the slurry, gas evolves and collects beneath the cover, **causing the cover to gradually rise**. The elevation of the floating cover provides an immediate physical indicator of internal gas pressure.
- **Gas Distribution Conduit:** The gas cover is fitted with an upper delivery valve coupled to distribution pipelines that deliver biogas directly to rural households for smokeless cooking and domestic illumination.
- **Spent Slurry Outlet:** Displaced digested residue discharges through an overflow outlet. This **spent slurry** is depleted of fermentable volatile solids but enriched in mineralized nitrogen, phosphorus, and potassium, serving as an exceptional organic manure/fertilizer.
- **Indian Pioneering Agencies:** The practical engineering and dissemination of biogas technology in India was spearheaded through the collaborative research and field extension initiatives of two major institutions:
  1. **Indian Agricultural Research Institute (IARI)**
  2. **Khadi and Village Industries Commission (KVIC)**

6. Microbes as Biocontrol Agents

Biocontrol refers to the deliberate utilization of biological organisms and natural ecological relationships (predation, parasitism, and pathogenesis) to regulate populations of agricultural insect pests, phytopathogenic fungi, and weeds, thereby minimizing reliance on toxic synthetic agrochemicals.

6.1 Principles of Biological Pest Suppression

Conventional synthetic insecticides and chemical pesticides exhibit severe ecological drawbacks: they are broad-spectrum poisons that accumulate across food webs (biomagnification), contaminate groundwater and surface runoffs, degrade soil microflora, and indiscriminately destroy beneficial insects along with targeted pests.

In contrast, **organic farming** and modern **Integrated Pest Management (IPM)** operate on a distinct holistic ecological paradigm:

- **Conservation of Biodiversity:** Biodiversity enhances ecological stability. A vibrant field landscape featuring diverse flora and fauna maintains intrinsic checks and balances.
- **Management vs Eradication:** The organic agriculturalist avoids total extermination of organisms labeled as "pests". Total eradication is ecologically counterproductive: without a baseline population of pest organisms, obligate predatory and beneficial parasitic insects that rely on them for nourishment and host oviposition cannot sustain viable populations. Instead, pests are stabilized at manageable, sub-economic injury thresholds.

6.2 Representative Biocontrol Agents

Biocontrol Organism	Biological Nature	Target Agricultural Pest / Pathogen	Mode of Action & Ecological Characteristics
Ladybird (Ladybird beetle)	Predatory beetle (conspicuous red and black markings)	**Aphids**	Active predatory consumption of sap-sucking nymphs and adults on crop foliage.
Dragonflies	Predatory insect (Odonata)	**Mosquitoes** (larvae and winged adults)	Aerial predation of mosquito adults and aquatic predation of larvae in wetland ecosystems.
Bacillus thuringiensis (Bt)	Spore-forming soil bacterium	**Butterfly caterpillars** and larval forms of vulnerable insect pests	Formulated as dry bacterial spores in water-dispersible sachets. Sprayed onto vulnerable crops (brassicas, fruit trees). When ingested by chewing larvae, protoxin crystals dissolve in the **alkaline gut**, releasing an active endotoxin that binds to midgut epithelial membranes, creates lytic pores, and kills the larvae while leaving non-target insects unharmed. Transgenic **Bt-cotton** expresses these insecticidal crystal genes endogenously.
***Trichoderma* species**	Free-living microscopic fungus	Diverse **soil-borne phytopathogenic fungi**	Highly abundant in plant **root ecosystems (rhizosphere)**. Acts through mycoparasitism, competitive exclusion, and antibiosis to protect root tissues from pathogenic wilts, rots, and damping-off fungi.
Baculoviruses (predominantly genus *Nucleopolyhedrovirus*)	Double-stranded DNA viral pathogens	**Insects and other arthropod pests**	Pathogens characterized by **species-specific, narrow-spectrum insecticidal activity**. Crucially, they exhibit **no negative impacts** on plants, mammals, birds, fish, or non-target beneficial insects (pollinators like honeybees). Indispensable tools in **Integrated Pest Management (IPM)** programs deployed across ecologically sensitive biotopes.

7. Microbes as Biofertilisers

The indiscriminate overuse of synthetic chemical fertilizers (NPK complexes) has triggered severe environmental pollution: progressive soil acidification, loss of native soil organic structure, and eutrophication of surrounding water bodies. Consequently, sustainable agricultural productivity demands a widespread transition toward **organic farming** through the application of **biofertilisers**.

Biofertilisers are defined as preparations containing living or latent cells of beneficial microorganisms (bacteria, fungi, and cyanobacteria) that colonize the rhizosphere or interior of host plants and enrich the nutrient capital of the soil by fixing atmospheric nitrogen, solubilizing minerals, or stimulating nutrient uptake.

7.1 Nitrogen-Fixing Bacteria

- **Symbiotic Nitrogen Fixers:**
  The rod-shaped bacterium ***Rhizobium*** establishes endosymbiotic associations within specialized root nodules of leguminous plants (family Fabaceae, e.g., peas, gram, beans, lentils). Inside the anaerobic nodule core (maintained by leghemoglobin), the bacterial enzyme *nitrogenase* reduces inert atmospheric gaseous nitrogen (N_2) into biologically available organic nitrogenous compounds (amino acids, amides) that are supplied directly to the host plant in return for photosynthetically fixed carbohydrates.

- **Free-Living (Non-Symbiotic) Nitrogen Fixers:**
  Certain bacteria reside as free-living organisms in soil and fix elemental dinitrogen without establishing obligate plant symbioses, thereby enriching the soil nitrogen pool. Notable examples highlighted in the syllabus include:
  - ***Azotobacter*** (strictly aerobic, free-living soil bacterium).
  - ***Azospirillum*** (associative/free-living microaerophilic bacterium commonly colonizing rhizosphere soils of cereal grasses).

7.2 Fungal Symbionts: Mycorrhizae

Certain soil fungi establish symbiotic associations with plant root systems, termed **mycorrhiza** (literally "fungus root"). Members of the zygomycetous/glomeromycetous genus ***Glomus*** are the most widespread formers of mycorrhizal associations.

- **Mineral Nutrition (Phosphorus Uptake):** The extensive external fungal hyphal network acts as an auxiliary absorbing root system, reaching deep into micro-soil pores to absorb immobile mineral nutrients—most notably **phosphorus** (as orthophosphate ions)—from the soil and translocate them efficiently into the plant tissues.
- **Holistic Secondary Benefits:** In addition to phosphorus delivery, plants colonized by mycorrhizal symbionts exhibit marked physiological advantages:
  1. Enhanced **resistance to soil-borne and root-borne fungal pathogens**.
  2. Elevated **tolerance to abiotic stress, particularly environmental salinity and drought**.
  3. Comprehensive promotion of overall vegetative vigor, root architecture, and shoot growth.
- **Symbiotic Reciprocity:** In return for mineral nutrients and water, the mycorrhizal fungus obtains essential photosynthetic assimilates (hexose sugars) and shelter from the host plant.

7.3 Cyanobacteria (Blue-Green Algae)

Cyanobacteria are photosynthetic, autotrophic prokaryotic microorganisms widely distributed across diverse aquatic (freshwater and marine) and terrestrial habitats. Many genera possess specialized, thick-walled, oxygen-impermeable cells termed *heterocysts* that house the oxygen-sensitive enzyme nitrogenase, enabling them to fix atmospheric nitrogen:

- **Major Genera:** *Anabaena*, *Nostoc*, and *Oscillatoria*.
- **Agronomic Role in Paddy Wetlands:** In flooded **paddy (rice) fields**, cyanobacteria function as indispensable biofertilisers, continuously enriching the aquatic substrate with fixed nitrogen.
- **Addition of Organic Matter:** In addition to biological nitrogen fixation, blue-green algae continuously shed mucilaginous cellular biomass upon death, contributing substantial quantities of **organic matter (humus)** to the soil and improving its water-retention capacity and physical tilth.

8. Worked Examples & Analytical Case Scenarios

Worked Example 1: Deducing Water Quality from BOD Laboratory Data

Problem: Three separate water samples—collected from (i) an unpolluted natural river, (ii) untreated domestic municipal sewage, and (iii) the clarified secondary effluent discharging from an operational municipal Sewage Treatment Plant—were subjected to a standard 5-day Biochemical Oxygen Demand (BOD) assay. The technician recorded the BOD values as:

- Sample X: $20\ \mathrm{mg/L}$
- Sample Y: $8\ \mathrm{mg/L}$
- Sample Z: $400\ \mathrm{mg/L}$

Identify which sample represents the highest polluting hazard and correctly correlate each sample to its source.

Analytical Solution:

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1. **Underlying Principle:** BOD measures the quantity of molecular oxygen consumed by heterotrophic bacteria while oxidising biodegradable organic matter in a liter of water. Thus, BOD directly and quantitatively reflects organic pollutant concentration. A higher BOD correlates with severe oxygen depletion and heightened ecological toxicity.
2. **Sample Z ($400\ \mathrm{mg/L}$):** Exhibits the highest BOD by an order of magnitude. This tremendous oxygen consumption reflects raw, unclarified human fecal matter and organic residues. *Conclusion: Sample Z is **Untreated Domestic Sewage** (Most Polluted).*
3. **Sample Y ($8\ \mathrm{mg/L}$):** Exhibits the lowest oxygen demand. A BOD below $10\ \mathrm{mg/L}$ is characteristic of well-aerated, clean natural lotic waters containing minimal putrescible substrates. *Conclusion: Sample Y is the **Clean River Water**.*
4. **Sample X ($20\ \mathrm{mg/L}$):** Reflects an intermediate value. It has undergone extensive biological oxidation in secondary aeration tanks (dropping from several hundred $\mathrm{mg/L}$ down to $20\ \mathrm{mg/L}$), but retains trace residual organic molecules. *Conclusion: Sample X is the **Secondary STP Effluent**.*

Worked Example 2: Enzyme Inhibition Kinetics of Statins

Problem: Explain the exact biochemical mechanism by which *statins*, isolated from the yeast *Monascus purpureus*, achieve clinical efficacy in managing coronary artery disease and hypercholesterolemia.

Analytical Solution:

1. **Source:** Statins are secondary metabolites synthesized during fermentation by the yeast *Monascus purpureus*.
2. **Target Enzyme:** Endogenous cellular biosynthesis of cholesterol occurs primarily in hepatocytes, with the committed, rate-limiting step catalyzed by the enzyme **HMG-CoA reductase** (β-hydroxy-β-methylglutaryl-coenzyme A reductase).
3. **Mechanism of Action:** Statins possess a chemical pharmacophore that structurally mimics the natural substrate (HMG-CoA). Consequently, statin molecules bind reversibly to the active catalytic site of HMG-CoA reductase. This represents classic **competitive enzyme inhibition**. By blocking the catalytic pocket, statins halt de novo cholesterol synthesis in liver cells, clearing circulating LDL-cholesterol from blood vessels.

9. NEET UG Exam-Style Practice Questions

The following original questions are formatted strictly according to the latest NEET UG examination blueprint, incorporating single-correct MCQs, Statement I & II assessments, Assertion-Reason pairs, and Matching Matrix items. Complete worked solutions accompany every question.

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Practice Question 1

Match the Columns

Match the microbially synthesized products listed in **Column I** with their producing microorganisms in **Column II** and select the correct option:

Column I (Product / Molecule)	Column II (Producing Microorganism)
A. Citric acid	1. *Monascus purpureus*
B. Cyclosporin A	2. *Aspergillus niger*
C. Statins	3. *Clostridium butylicum*
D. Butyric acid	4. *Trichoderma polysporum*

(A) A-2, B-4, C-1, D-3

(B) A-2, B-1, C-4, D-3

(C) A-3, B-4, C-1, D-2

(D) A-4, B-2, C-3, D-1

Correct Option: (A) [A-2, B-4, C-1, D-3]

Detailed Explanation:

- **A. Citric acid** is an organic acid synthesized industrially by the fungus *Aspergillus niger* (A → 2).
- **B. Cyclosporin A** is an immunosuppressive oligopeptide produced by the filamentous fungus *Trichoderma polysporum* (B → 4).
- **C. Statins** are blood-cholesterol lowering molecules synthesized by the yeast *Monascus purpureus* (C → 1).
- **D. Butyric acid** is generated by the anaerobic bacterium *Clostridium butylicum* (D → 3).

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Practice Question 2

Two-Statement Evaluation

Given below are two statements:

Statement I: Conversion of fresh milk into curd by Lactic Acid Bacteria (LAB) improves its nutritional quality by increasing the concentration of vitamin B_{12}.

Statement II: In the human stomach, Lactic Acid Bacteria induce the coagulation of milk proteins and act as opportunistic disease-causing agents.

In light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Option: (C) [Statement I is correct but Statement II is incorrect]

Detailed Explanation:

Statement I is entirely accurate: during milk fermentation, proliferating LAB synthesize and enrich curd with vitamin B_{12}. Statement II is false: in our stomach, LAB do not act as opportunistic pathogens; rather, they play a very beneficial protective probiotic role by actively checking and suppressing harmful disease-causing microbes.

Practice Question 3

Conceptual MCQ

Which of the following alcoholic beverages are manufactured by fermenting nutrient broths **without** subjecting the mixture to downstream distillation?

(A) Whisky and Brandy

(B) Wine and Beer

(C) Rum and Whisky

(D) Beer and Rum

Correct Option: (B) [Wine and Beer]

Detailed Explanation:

According to syllabus specifications, fermented beverages are categorized based on distillation processing: **Wine and beer are produced without distillation** (yielding lower alcohol percentages). In contrast, whisky, brandy, and rum are produced by distillation of the fermented broth to concentrate the alcohol content.

BOSONANCE

Practice Question 4

Assertion - Reason

Given below are two statements: an **Assertion (A)** and a **Reason (R)**.

Assertion (A): Commercial bottled fruit juices purchased from market outlets are noticeably clearer and more transparent than freshly prepared homemade fruit juices.

Reason (R): Commercial bottled fruit juices are routinely clarified during processing through the enzymatic action of pectinases and proteases.

Select the correct answer from the choices below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Option: (A) [Both (A) and (R) are true and (R) is the correct explanation of (A)]

Detailed Explanation:

Homemade fruit juices remain turbid due to suspended insoluble pectins, cellular debris, and protein-tannin colloids. Industrial processing incorporates microbial pectinases and proteases, which enzymatically degrade structural pectins and proteins, precipitating cloud-forming colloids and producing a clear, transparent beverage. Both statements are true and Reason (R) directly explains Assertion (A).

Practice Question 5

Single-Correct MCQ

A patient admitted to an intensive cardiac care unit has suffered an acute myocardial infarction caused by a coronary thrombus. Which of the following microbially derived enzyme preparations is administered intravenously as a specific "clot buster"?

(A) Cyclosporin A obtained from *Trichoderma polysporum*

(B) Statin obtained from *Monascus purpureus*

(C) Streptokinase obtained from *Streptococcus* and modified by genetic engineering

(D) Amylase obtained from *Aspergillus niger*

Correct Option: (C) [Streptokinase obtained from *Streptococcus* and modified by genetic engineering]

Detailed Explanation:

Streptokinase, synthesized by the bacterium *Streptococcus* and optimized via genetic engineering, functions as a potent thrombolytic agent ("clot buster"). It dissolves fibrin clots within occluded coronary blood vessels, restoring blood flow to ischemic myocardium in heart attack patients.

BOSONANCE

Practice Question 6

Process Sequencing / Concept MCQ

During secondary (biological) sewage treatment, what happens immediately **after** the Biochemical Oxygen Demand (BOD) of the sewage effluent has been significantly reduced in the aeration tank?

(A) The effluent is immediately pumped into an anaerobic sludge digester to produce methane.

(B) The effluent is passed into a settling tank where microbial flocs are allowed to sediment into activated sludge.

(C) The effluent is subjected to sequential filtration to remove grit and small pebbles.

(D) Air is pumped into the settling tank to kill anaerobic bacteria.

Correct Option: (B) [The effluent is passed into a settling tank where microbial flocs are allowed to sediment into activated sludge]

Detailed Explanation:

Once aerobic respiration by flocs reduces the BOD significantly, the mixed liquor leaves the aeration chamber and enters the secondary settling tank. In this tank, agitation ceases, allowing the bacterial-fungal flocs to sediment by gravity. This sediment is termed **activated sludge**. A small portion of this sludge returns to the aeration tank as an inoculum, while the bulk is directed to the anaerobic sludge digester.

BOSONANCE

Practice Question 7

Two-Statement Evaluation

Given below are two statements regarding Biochemical Oxygen Demand (BOD):

Statement I: BOD test measures the rate of oxygen uptake by microorganisms in a water sample and thus serves as an indirect measure of the organic matter present.

Statement II: Highly polluted municipal sewage exhibits an exceptionally low BOD value because microbes fail to survive in toxic wastewater.

In light of the above statements, choose the correct answer:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Correct Option: (C) [Statement I is correct but Statement II is incorrect]

Detailed Explanation:

Statement I is true: the BOD test determines oxygen uptake during the microbial oxidation of organic matter, providing an indirect measure of organic pollution. Statement II is false: heavily polluted sewage contains massive amounts of biodegradable organic matter, driving intense microbial respiration and resulting in an **exceptionally high BOD**, not a low BOD.

Practice Question 8

Single-Correct MCQ

The inflammable gaseous mixture generated inside the anaerobic sludge digester of a sewage treatment plant consists predominantly of:

(A) Methane, Hydrogen sulphide, and Carbon dioxide

(B) Methane, Oxygen, and Carbon monoxide

(C) Carbon dioxide, Nitrogen dioxide, and Chlorine

(D) Hydrogen, Ethane, and Sulphur dioxide

Correct Option: (A) [Methane, Hydrogen sulphide, and Carbon dioxide]

Detailed Explanation:

In anaerobic sludge digesters, anaerobic bacteria (methanogens) digest the bacterial and fungal biomass of the flocs. During this digestion, the microbes synthesize a mixture of gases consisting of **methane (**$\mathrm{CH_4}$**), hydrogen sulphide (**$\mathrm{H_2S}$**), and carbon dioxide (**$\mathrm{CO_2}$**)**. This mixture constitutes biogas.

BOSONANCE

Practice Question 9

Single-Correct MCQ

Methanogenic bacteria like *Methanobacterium* are **NOT** typically encountered in which of the following ecological habitats?

(A) Anaerobic sludge of a sewage treatment plant

(B) The rumen of herbivorous cattle

(C) Mechanically agitated aeration tanks of a sewage treatment plant

(D) Flooded, anoxic bottom mud of a rural biogas digester

Correct Option: (C) [Mechanically agitated aeration tanks of a sewage treatment plant]

Detailed Explanation:

Methanogens are **obligate/facultative anaerobes** that thrive in strictly oxygen-deficient environments (rumen of cattle, anaerobic sludge digesters, flooded biogas pits). The aeration tank of an STP is continuously injected with air and agitated to sustain **aerobic flocs**; methanogens cannot function under these highly oxygenated conditions.

BOSONANCE

Practice Question 10

Match the Columns

Match the biocontrol agents in **Column I** with their targeted agricultural pests in **Column II**:

Column I (Biocontrol Agent)	Column II (Targeted Organism / Pest)
A. Ladybird beetle	1. Butterfly caterpillars
B. Dragonfly	2. Plant root pathogens
C. *Bacillus thuringiensis*	3. Mosquitoes
D. *Trichoderma* species	4. Aphids

(A) A-4, B-3, C-1, D-2

(B) A-3, B-4, C-1, D-2

(C) A-4, B-1, C-3, D-2

(D) A-2, B-3, C-4, D-1

Correct Option: (A) [A-4, B-3, C-1, D-2]

Detailed Explanation:

- **Ladybird beetle** (red with black spots) feeds upon **aphids** (A → 4).
- **Dragonflies** prey upon **mosquitoes** (B → 3).
- ***Bacillus thuringiensis*** (Bt) produces endotoxins targeting **butterfly caterpillars** (C → 1).
- ***Trichoderma*** is a free-living root fungus used against **plant root pathogens** (D → 2).

BOSONANCE

Practice Question 11

Multi-Statement Selection

Read the following statements regarding the viral biocontrol agents belonging to the genus *Nucleopolyhedrovirus* (Baculoviruses):

1. They attack insects and other arthropods.
2. They are excellent candidates for broad-spectrum, non-specific chemical pesticide replacement.
3. They exhibit no negative impacts on plants, mammals, birds, fish, or non-target insects.
4. They are especially desirable when beneficial insects are being conserved in an Integrated Pest Management (IPM) program.

Which of the above statements are **correct**?

(A) 1 and 2 only

(B) 1, 3, and 4 only

(C) 2, 3, and 4 only

(D) 1, 2, 3, and 4

Correct Option: (B) [1, 3, and 4 only]

Detailed Explanation:

Statement 2 is incorrect because baculoviruses are **species-specific, narrow-spectrum** insecticidal agents, NOT broad-spectrum agents. Statements 1, 3, and 4 are completely correct factual statements verbatim from the syllabus regarding *Nucleopolyhedrovirus*.

BOSONANCE

Practice Question 12

Single-Correct MCQ

A farmer observes that several fungal biofertilisers forming mycorrhizal symbioses belong to the genus *Glomus*. Which primary mineral nutrient is absorbed from the soil and translocated to the host plant by this fungal symbiont?

(A) Nitrogen

(B) Phosphorus

(C) Potassium

(D) Iron

Correct Option: (B) [Phosphorus]

Detailed Explanation:

Mycorrhizal fungi of the genus *Glomus* absorb **phosphorus** from the soil and deliver it to the host plant. Additionally, the plant gains enhanced resistance to root-borne pathogens, tolerance to salinity and drought, and overall developmental vigor.

Practice Question 13

Assertion - Reason

Given below are two statements: an **Assertion (A)** and a **Reason (R)**.

Assertion (A): Cyanobacteria such as *Anabaena*, *Nostoc*, and *Oscillatoria* serve as vital biofertilisers in flooded paddy fields.

Reason (R): Cyanobacteria are autotrophic, photosynthetic microbes that can fix atmospheric nitrogen and add organic matter to the soil.

Select the correct answer from the choices below:

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Correct Option: (A) [Both (A) and (R) are true and (R) is the correct explanation of (A)]

Detailed Explanation:

Both statements are true. Cyanobacteria thrive in the warm, illuminated aquatic environment of flooded rice paddies. Because they are autotrophic and fix atmospheric nitrogen into organic forms while shedding biomass (humus), they significantly enrich soil fertility without chemical nitrogen fertilizers. Thus, (R) provides the direct scientific rationale for (A).

BOSONANCE

Practice Question 14

Identification MCQ

Which of the following pairings of a nitrogen-fixing microorganism with its mode of existence is **incorrectly** matched?

(A) *Rhizobium* — Symbiotic in root nodules of leguminous plants

(B) *Azotobacter* — Free-living nitrogen-fixing bacterium in soil

(C) *Azospirillum* — Free-living nitrogen-fixing bacterium in soil

(D) *Glomus* — Free-living nitrogen-fixing cyanobacterium in water

Correct Option: (D) [*Glomus* — Free-living nitrogen-fixing cyanobacterium in water]

Detailed Explanation:

Pairing (D) is completely incorrect: *Glomus* is a **fungus** that forms symbiotic **mycorrhizal associations** with plant roots for **phosphorus** uptake. It is neither a cyanobacterium nor an organism that fixes nitrogen. Options (A), (B), and (C) are all accurate pairings according to the syllabus.

Practice Question 15

Single-Correct MCQ

Large macroscopic holes in 'Swiss cheese' are generated as a result of:

(A) Synthesis of methane gas by *Methanobacterium*

(B) Release of large amounts of CO_2 by the bacterium *Propionibacterium sharmani*

(C) Lysis of casein curd by the fungal mould *Penicillium roqueforti*

(D) Vigorous mechanical aeration during milk coagulation

Correct Option: (B) [Release of large amounts of CO_2 by the bacterium *Propionibacterium sharmani*]

Detailed Explanation:

The distinctive large holes (eyes) in Swiss cheese develop because the fermenting bacterium *Propionibacterium freudenreichii* subsp. *shermanii* (referred to as *Propionibacterium sharmani*) synthesizes copious amounts of **carbon dioxide** (CO_2) gas during ripening.

10. Common Pitfalls & NEET High-Yield Revision Checklist

Frequent NEET Confusion Points & Traps

- **Distillation Status of Alcoholic Drinks:**
  • *Without Distillation:* Wine and Beer (Lower alcohol concentration).
  • *With Distillation:* Whisky, Brandy, and Rum (Concentrated alcohol).

BOSONANCE

- ***Trichoderma* Species Dual Roles:**
  • *Trichoderma polysporum* is an industrial fungus yielding the immunosuppressive bioactive drug **Cyclosporin A**.
  • Free-living *Trichoderma* species inhabiting root ecosystems act as **biocontrol agents** against root fungal pathogens.
- **Taxonomic Identities of Bioactive Producers:**
  • *Monascus purpureus* is a **yeast** (fungus), NOT a bacterium. It synthesizes statins.
  • *Streptococcus* is a **bacterium**. It produces streptokinase.
  • *Aspergillus niger* is a **fungus**. It produces citric acid.
  • *Acetobacter aceti* and *Clostridium butylicum* are **bacteria**.
- **Primary vs Secondary Sludge:**
  • **Primary Sludge:** Physically settled grit and solid debris; formed in primary settling tanks; strictly non-biological.
  • **Activated Sludge:** Biologically formed sediment composed of living and dead flocs (aerobic bacteria + fungal mycelia); formed in secondary settling tanks.
- **Baculoviruses / Nucleopolyhedrovirus:**
  They have **narrow-spectrum, species-specific** insecticidal activity. Questions frequently insert "broad-spectrum" as an incorrect distractor. They do not harm non-target insects, mammals, birds, or plants.
- **Gases in Anaerobic Sludge Digester vs Rumen Biogas:**
  • Anaerobic Sludge Digester: $\mathrm{CH_4} + \mathrm{H_2S} + \mathrm{CO_2}$
  • Cattle Dung / Rumen Biogas: Predominantly $\mathrm{CH_4}$ accompanied by $\mathrm{CO_2}$ and $\mathrm{H_2}$.

Rapid Revision Checklist

Topic	Key Organism / Concept	Critical Fact for NEET
Curd	*Lactobacillus* / LAB	Coagulates milk proteins; elevates **vitamin B_{12}**; checks enteric pathogens.
Bread / Beer	*Saccharomyces cerevisiae*	Baker's yeast / Brewer's yeast; leavens dough via $\mathrm{CO_2}$; ferments maltose to ethanol.
Swiss Cheese	*Propionibacterium sharmani*	Bacterium; produces large gas holes due to copious $\mathrm{CO_2}$.
Roquefort Cheese	*Penicillium roqueforti*	Fungus; ripened by mould growth for specific flavor.
Penicillin	*Penicillium notatum*	Discovered by Fleming (chance observation on *Staphylococci*); therapeutic potential established by Florey and Chain (Nobel Prize 1945).
Clot Buster	*Streptococcus* (Streptokinase)	Enzyme modified by genetic engineering; clears blood clots in myocardial infarction.
Organ Transplant	*Trichoderma polysporum*	Yields Cyclosporin A (immunosuppressive drug).
Cholesterol Lowering	*Monascus purpureus*	Yields statins; competitively inhibits HMG-CoA reductase (cholesterol synthesis enzyme).
Sewage Flocs	Bacteria + Fungal filaments	Aerobic masses in aeration tank; drastically reduce BOD.
Biogas Plant	*Methanobacterium* (Methanogens)	Anaerobic digester; 10–15 ft concrete tank with floating gas cover; developed by IARI and KVIC.
Biocontrol (Insects)	Ladybird & Dragonfly	Ladybird controls aphids; Dragonflies control mosquitoes.
Biocontrol (Caterpillars)	*Bacillus thuringiensis*	Spore formulations; toxic crystal proteins dissolve in alkaline insect gut.
IPM Biocontrol	*Baculovirus* (*Nucleopolyhedrovirus*)	Species-specific, narrow-spectrum; leaves beneficial non-target organisms safe.
Biofertiliser (P)	*Glomus* (Mycorrhiza)	Absorbs phosphorus; enhances pathogen resistance and drought/salinity tolerance.

BOSONANCE

Topic	Key Organism / Concept	Critical Fact for NEET
Biofertiliser (N)	*Rhizobium*, *Azotobacter*, *Azospirillum*	*Rhizobium* is symbiotic in legume nodules; *Azotobacter* and *Azospirillum* are free-living.
Paddy Biofertiliser	*Anabaena*, *Nostoc*, *Oscillatoria*	Photosynthetic cyanobacteria; fix nitrogen and add organic matter to rice fields.

11. Final Takeaways

Microorganisms represent indispensable biological assets across domestic, clinical, industrial, and agricultural domains. From everyday dietary enrichments (LAB, baker's yeast, artisanal cheeses) to life-saving clinical agents (penicillin, streptokinase, cyclosporin A, statins), heterotrophic and autotrophic microbes sustain critical human welfare systems. In environmental management, microbial aeration and anaerobic digestion treat millions of gallons of municipal wastewater, while methanogenic digestion of cattle dung delivers clean, renewable biogas to rural communities. In agriculture, transitioning from hazardous synthetic chemicals to microbial biocontrol agents (*Bt*, *Trichoderma*, baculoviruses) and biofertilisers (*Rhizobium*, *Glomus*, cyanobacteria) forms the foundation of sustainable ecological farming.

BOSONANCE